

RAJVEER SINGH

AI Engineer | Full-Stack AI Application Developer

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SUMMARY

AI Engineer and full-stack developer with 3+ years of experience building and shipping production-grade, AI-powered applications. Specialized in integrating ML and LLM systems into scalable web platforms—RAG pipelines, model-serving APIs, real-time data processing, and 3D/point-cloud analytics. Combines hands-on machine-learning integration (PyTorch, Hugging Face, FastAPI) with strong end-to-end engineering across the modern web stack, enabling AI features to move from prototype to reliable, deployed product.

TECHNICAL SKILLS

AI / ML: PyTorch, scikit-learn, Hugging Face Transformers, model inference & serving, feature pipelines, model evaluation, computer vision (3D point-cloud / LiDAR)

LLMs & GenAI: OpenAI & Anthropic APIs, LangChain / LlamaIndex, Retrieval-Augmented Generation (RAG), embeddings, vector databases (pgvector, Pinecone, FAISS), prompt engineering, AI agents

Languages: Python, TypeScript, JavaScript (ES6+), SQL

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, WebSockets / Socket.IO, MQTT, authentication & authorization

Data: PostgreSQL / pgvector, MongoDB, pandas, NumPy, data ingestion & ETL pipelines

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, Material UI, React Flow, data visualization

Cloud & MLOps: Azure App Services, Azure Blob Storage, AWS S3, AWS Cognito, Azure DevOps, GitHub Actions / CI-CD, Docker

Specializations: AI application engineering, RAG systems, real-time data platforms, IoT & BLE analytics, computer-vision pipelines

WORK EXPERIENCE

AI Engineer / Full-Stack Developer | Celebal Technologies, Jaipur

Jul 2024 – Present

- Build AI-powered enterprise applications end to end, integrating ML/LLM models into production web platforms using Python (FastAPI), Next.js, Node.js, and TypeScript.
- Engineered an AI analytics platform for mining safety—serving deep-learning anomaly-detection models over large-scale 3D LiDAR point-cloud data with secure, scalable ingestion pipelines.
- Designed and deployed model-serving REST APIs and inference workflows, focusing on latency, throughput, security, and maintainability.
- Developed real-time data-processing and visualization features (LiDAR point clouds, BLE telemetry, live conversation analytics) for high-frequency streaming workloads.
- Collaborated with clients, product, and data teams to translate business problems into AI/ML solutions, delivering demos and driving production releases across multiple projects.

Full-Stack Developer | Intourist, Jaipur

Apr 2023 – Jun 2024

- Developed and maintained web applications and travel-portal solutions using PHP, WordPress, HTML, and CSS, building responsive UIs and custom features to client requirements.
- Integrated third-party APIs and backend services to power booking workflows, content management, and customer-facing functionality.
- Contributed to performance optimization, testing, bug fixing, and production deployments across multiple client projects.

PROJECTS

AI-Powered Mining LiDAR Analytics Platform | *Next.js, FastAPI, PyTorch, AWS S3/Cognito, pgvector*

- Built a full-stack AI platform that processes, visualizes, and reports on large-scale LiDAR scans, applying deep-learning models for mining bolt-anomaly detection.
- Implemented model-serving inference APIs in FastAPI and secure, scalable file-ingestion pipelines (AWS S3 + Cognito) for high-volume point-cloud datasets.
- Developed interactive 3D point-cloud visualization and automated reporting to support engineering analysis and operational decision-making.

RAG-Based Document Intelligence Assistant | *Python, LangChain, OpenAI/Anthropic APIs, pgvector, FastAPI*

- Designed a Retrieval-Augmented Generation assistant that answers natural-language queries over private document collections using embeddings and a vector database.
- Implemented chunking, embedding, semantic retrieval, and prompt-orchestration pipelines, with source-grounded responses to reduce hallucination.
- Exposed the system through a FastAPI backend and a React chat interface with streaming responses.

Real-Time Contact Center Intelligence Platform | *React, SignalR, TypeScript, NLP*

- Built real-time agent-assist dashboards using SignalR live data streaming, surfacing call insights, sentiment indicators, and conversation metrics powered by NLP.
- Optimized frontend performance for high-frequency real-time updates while maintaining a responsive user experience.

IoT Device Monitoring & Analytics Platform | *React, Node.js, BLE, MQTT, time-series analytics*

- Developed a platform for ingesting, monitoring, and visualizing telemetry from IoT and BLE-enabled devices, with real-time dashboards and anomaly-aware reporting modules.
- Integrated BLE communication and device-configuration workflows for seamless interaction with field-deployed hardware.

Transposer – Chrome Extension | *React, Web Audio API*

chromewebstore.google.com/detail/transposer-transpose-pitc/bhlokigpgcdidlakfodbghhgbdpgaobb

- Published a Chrome extension for real-time audio pitch shifting, tempo control, and equalizer adjustment using low-latency browser audio processing.
- Achieved 4,000+ weekly active users on the Chrome Web Store with a 4.5+ rating.

EDUCATION

Master of Computer Applications (MCA)

2022 – 2024

Rajasthan Technical University, Kota

ACHIEVEMENTS

- Winner – Rajasthan Digifest × TIE Global Hackathon: built a Smart Canal Irrigation Monitoring & Management System using IoT sensors, LoRa, React, Node.js, MongoDB, GeoJSON, and real-time data streaming.
- Winner – AI + Data + CRM Summit Hackathon (Wipro): built an AI-driven solution focused on operational efficiency and data intelligence.
- Winner – ISIM Hackathon 2.0: created an IoT-enabled Smart Parking System (ESP32, real-time occupancy monitoring, mobile booking, web dashboards).
- First Runner-Up – HACK JKLU 3.0, a 36-hour competitive innovation hackathon.
- Third Position – IoT-Based Hackathon organized by AIET Jaipur.